

Ans. to the ques. no-01

Under static scoping (lexical scoping), the scope of a variable is determined entirely at compile-time based on its physical location in the code.

The program is written below $\Rightarrow$ (in C)

```
#include <stdio.h>

int a = 90;

void display-value()
{
    printf("The value, a: %d\n", a);
}

int main()
{
    int a = 56;
    printf("Inside the main function, a: %d\n", a);
    display-value();
    return 0;
}
```

static_scoping.c - Code::Blocks 25.03

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<global> display_value(): void

static_scoping.c X

```
1 //Aziza Zaman Sadia
2 #include <stdio.h>
3
4 int a = 90;
5
6 void display_value() {
7     printf("The value, a : %d\n", a);
8 }
9
10 int main() {
11     int a = 56;
12
13     printf("Inside the main function, a : %d\n", a);
14     display_value();
15
16     return 0;
17 }
18
```

C:\Users\Hp\Downloads\static_scoping.exe

```
Inside the main function, a : 56
The value, a : 90

Process returned 0 (0x0)   execution time : 0.052 s
Press any key to continue.
```

Logs & others

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File Line Message

==== Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ====

C:\Users\Hp\Downloads\static_scoping.c

C/C++

Windows (CR+LF)

WINDOWS-1252

Line 7, Col 27, Pos 108

Insert

Read/Write

default

abc

Ans. to the ques. no-02

Control Structures ⇒

At the hardware layer, execution is strictly sequential. Here, control structures are abstractions in high-level language design that allows to alter this flow.

Control structures might be divided into primarily three categories - Selection/Branching (if, switch), Looping (for, while, do-while), Jump (goto, continue, break)

Flow Control Statements ⇒

① If ⇒

```
if (condition)
{
    //statement-block (do after true)
}
```

② while ⇒

```
while (condition)
{
    //body
}
```

③ for ⇒

```
for (initialization; condition; loop-increment)
{
    // body
}
```

④ do-while ⇒

```
do
{
    // body
} while (condition);
```

⑤ if - else ⇒

```
if (condition)
{
    // body
}
else
{
    // body
}
```

⑥ switch (expression)

```
{
    case case-1:
        // statements
    case case-2:
        // statements
    default:
        // statements
}
```


Ans. to the ques. no-03

The program is written below $\Rightarrow$ (in C)

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#define PI 3.1416
```

```
void calculate_area_perimeter (int radius, double *area,  
                                double *perimeter)
```

```
{  
    double r = (double) radius;  
    *area = PI * r * r;  
    *perimeter = 2 * PI * r;  
}
```

```
int main()
```

```
{  
    int row = 5;  
    int col = 5;  
    int radius_matrix [row] [col];  
    double area_matrix [row] [col];
```

```
    printf ("Enter the values of radius for %d x %d matrix,"  
            row, col);
```

```
    for (int i = 0; i < row; i++)
```

```
    {  
        for (int j = 0; j < col; j++)
```

```
        {  
            printf ("Element [%d] [%d] : ", i, j);
```

```
            scanf ("%d", &radius_matrix [i] [j]);
```

```
        }  
    }
```

```

FILE *file_ptr = fopen("circle-metrics.txt", "a");

if (file_ptr == NULL)
{
    printf("Error!");
    return 1;
}

fprintf(file_ptr, "%-12s %-12s %-18s %-18s\n",
        "Serial No.", "Radius", "Area",
        "Perimeter");

fprintf(file_ptr, "\n");

int serial_no = 1;
for (int i = 0; i < row; i++)
{
    for (int j = 0; j < col; j++)
    {
        int current_radius = radius_matrix[i][j];
        double area_calc = 0.0;
        double perimeter_calc = 0.0;

        calculate_area_perimeter(current_radius,
                                &area_calc, &perimeter_calc);

        area_matrix[i][j] = area_calc;

        fprintf(file_ptr, "%-12d %-12d %-18f %-18f\n",
                serial_no, current_radius, area_calc,
                perimeter_calc);

        serial_no++;
    }
}

fclose(file_ptr);

```

```

printf ("InRadius Matrix : \n");
for (int i=0 ; i<row ; i++)
{
    for (int j=0 ; j<col ; j++)
    {
        printf (" %.d " , radius_matrix[i][j]);
    }
    printf ("\n");
}
printf ("Area Matrix : \n");
for (int i=0 ; i<row ; i++)
{
    for (int j=0 ; j<col ; j++)
    {
        printf (" %.d " , area_matrix[i][j]);
    }
    printf ("\n");
}
return 0;
}

```


circle_problem.c - Code::Blocks 25.03

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circle_problem.c X

```
1 //Aziza Zaman Sadia
2 #include <stdio.h>
3 #include <stdlib.h>
4
5 #define PI 3.1416
6
7 void calculate_area_perimeter(int radius, double *area, double *perimeter) {
8     double r = (double)radius;
9     *area = PI * r * r;
10    *perimeter = 2.0 * PI * r;
11 }
12
13 int main() {
14     int row = 5;
15     int col = 5;
16     int radius_matrix[row][col];
17     double area_matrix[row][col];
18
19
20     printf("Enter the values of radius for %d x %d Matrix:\n", row, col);
21     for (int i = 0; i < row; i++) {
22         for (int j = 0; j < col; j++) {
23             printf("Element [%d][%d]: ", i, j);
24             scanf("%d", &radius_matrix[i][j]);
25         }
26     }
27
28     FILE *file_ptr = fopen("circle_metrics.txt", "a");
29     if (file_ptr == NULL) {
30         printf("Error!\n");
31         return 1;
32     }
33
34     fprintf(file_ptr, "%-12s %-12s %-18s %-18s\n", "Serial No.", "Radius", "Area", "Perimeter");
35     fprintf(file_ptr, "\n");
36
37     int serial_no = 1;
38     for (int i = 0; i < row; i++) {
39         for (int j = 0; j < col; j++) {
40             int current_radius = radius_matrix[i][j];
41
42             calculate_area_perimeter(current_radius, &area_matrix[i][j], &perimeter_matrix[i][j]);
43
44             serial_no++;
45         }
46     }
47 }
```

Logs & others

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File	Line	Message
		== Build file: "no target" in "no project" (compiler: unknown) ==
		== Build finished: 0 error(s), 0 warning(s) (0 minute(s), 0 second(s)) ==

C:\Users\Hp\Downloads\circle_problem.exe

Enter the values of radius for 5 x 5 Matrix:

Element [0][0]: 1
Element [0][1]: 2
Element [0][2]: 3
Element [0][3]: 4
Element [0][4]: 5
Element [1][0]: 6
Element [1][1]: 7
Element [1][2]: 8
Element [1][3]: 9
Element [1][4]: 10
Element [2][0]: 11
Element [2][1]: 12
Element [2][2]: 13
Element [2][3]: 14
Element [2][4]: 15
Element [3][0]: 16
Element [3][1]: 17
Element [3][2]: 18
Element [3][3]: 19
Element [3][4]: 20
Element [4][0]: 21
Element [4][1]: 22
Element [4][2]: 23
Element [4][3]: 24
Element [4][4]: 25

Radius Matrix

1 2 3 4 5
6 7 8 9 10
11 12 13 14 15
16 17 18 19 20
21 22 23 24 25

Area Matrix

3.14 12.57 28.27 50.27 78.54
113.10 153.94 201.06 254.47 314.16
380.13 452.39 530.93 615.75 706.86
804.25 907.92 1017.88 1134.12 1256.64
1385.45 1520.53 1661.91 1809.56 1963.50

Process returned 0 (0x0) execution time : 23.157 s
Press any key to continue.

files list X

circle_problem.c - Code::Blocks 25.03

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circle_problem.c X

```
31     return 1;
32 }
33
34 fprintf(file_ptr, "%-12s %-12s %-18s %-18s\n", "Serial No.", "Radius", "Area", "Perimeter");
35 fprintf(file_ptr, "\n");
36
37 int serial_no = 1;
38 for (int i = 0; i < row; i++) {
39     for (int j = 0; j < col; j++) {
40         int current_radius = radius_matrix[i][j];
41         double area_calc = 0.0;
42         double perimeter_calc = 0.0;
43
44         calculate_area_perimeter(current_radius, &area_calc, &perimeter_calc);
45
46         area_matrix[i][j] = area_calc;
47
48         fprintf(file_ptr, "%-12d %-12d %-18f %-18f\n", serial_no, current_radius, area_calc, perimeter_calc);
49
50         serial_no++;
51     }
52 }
53
54 fclose(file_ptr);
55
56 printf("\nRadius Matrix\n");
57 for (int i = 0; i < row; i++) {
58     for (int j = 0; j < col; j++) {
59         printf("%d ", radius_matrix[i][j]);
60     }
61     printf("\n");
62 }
63
64 printf("\nArea Matrix\n");
65 for (int i = 0; i < row; i++) {
66     for (int j = 0; j < col; j++) {
67         printf("%.2f ", area_matrix[i][j]);
68     }
69     printf("\n");
70 }
71
72 return 0;
73 }
74
75 }
```

Logs & others

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C:\Users\Hp

File Line Message

circle_metrics.txt

File Edit View H1 ...

Serial No.	Radius	Area	Perimeter
1	1	3.141600	6.283200
2	2	12.566400	12.566400
3	3	28.274400	18.849600
4	4	50.265600	25.132800
5	5	78.540000	31.416000
6	6	113.097600	37.699200
7	7	153.938400	43.982400
8	8	201.062400	50.265600
9	9	254.469600	56.548800
10	10	314.160000	62.832000
11	11	380.133600	69.115200
12	12	452.390400	75.398400
13	13	530.930400	81.681600
14	14	615.753600	87.964800
15	15	706.860000	94.248000
16	16	804.249600	100.531200
17	17	907.922400	106.814400
18	18	1017.878400	113.097600
19	19	1134.117600	119.380800
20	20	1256.640000	125.664000
21	21	1385.445600	131.947200
22	22	1520.534400	138.230400
23	23	1661.906400	144.513600
24	24	1809.561600	150.796800
25	25	1963.500000	157.080000

Serial No. Radius Area Perimeter

1	1	3.141600	6.283200
2	2	12.566400	12.566400
3	3	28.274400	18.849600
4	4	50.265600	25.132800
5	5	78.540000	31.416000
6	6	113.097600	37.699200
7	7	153.938400	43.982400
8	8	201.062400	50.265600
9	9	254.469600	56.548800
10	10	314.160000	62.832000
11	11	380.133600	69.115200
12	12	452.390400	75.398400
13	13	530.930400	81.681600
14	14	615.753600	87.964800
15	15	706.860000	94.248000
16	16	804.249600	100.531200
17	17	907.922400	106.814400
18	18	1017.878400	113.097600
19	19	1134.117600	119.380800
20	20	1256.640000	125.664000
21	21	1385.445600	131.947200
22	22	1520.534400	138.230400
23	23	1661.906400	144.513600
24	24	1809.561600	150.796800
25	25	1963.500000	157.080000

Ln 1, Col 1 3,330 chara Plain text 70% Windows (UTF-8

Closed files list X